Saturday — Python Cheat Sheet

The names, keywords, operators, and methods used in Saturday’s video/PDF-aligned lessons. This is a quick reference, not a replacement for topic explanations.

Displaying output

```
print("Hello")
print(5 + 3)
print("Name:", name)
```

Variables & basic types

```
name = "Ali"
age = 20
price = 12.5
is_present = True
```

Input & casting

```
name = input("Name: ")
age = int(input("Age: "))
price = float(input("Price: "))
```

Type inspection & conversions

```
type(3.5)
int("25")
float("4.5")
str(45)
bool(1)
```

String functions & methods

```
len("Python")
"hello".upper()
"HELLO".lower()
"Tony Stark".find("Stark")
"cat".replace("c", "b")
"a" in "cat"
```

Arithmetic & assignment

```
+  -  *  /  //  %  **
=
+=  -=  *=  /=
```

Comparison & logic

```
==  !=  >  <  >=  <=
and  or  not
```

Decisions

```
if marks >= 50:
    print("Pass")
else:
    print("Try again")
```

Loops

```
for i in range(1, 6):
    print(i)
while count < 5:
    count += 1
break  # stop
continue  # skip
```

Lists

```
fruits = ["apple", "banana"]
fruits.append("mango")
fruits.insert(0, "pear")
fruits[0]
fruits[0:2]
len(fruits)
```

Tuples & sets

```
scores = (90, 95, 90)
scores.count(90)
scores.index(95)
unique = {1, 1, 2}
unique.add(3)
empty_set = set()
```

Dictionaries

```
marks = {"Math": 90}
marks["Math"]
marks["English"] = 80
for subject in marks:
    print(subject, marks[subject])
```

Functions

```
def add(a, b):
    return a + b
print(add(2, 3))
min([2, 3])
max([2, 3])
sum([2, 3])
```

Modules

```
import math
print(math.sqrt(16))
print(math.pow(2, 3))
print(math.pi)
print("sqrt" in dir(math))
import random
print(random.randint(1, 10))
```

Remember

input() returns a string; convert before doing numeric calculations. Python indexing starts at zero. Indentation defines code blocks. Lists/dictionaries can change; tuples cannot. Use return when a function should give a value back.